

Determine Distance on the Coordinate Plane

Lesson 7-6

Name:

Date:

Class:



TODAY'S OBJECTIVES

Content Objective: I can find the distance between two points on the coordinate plane using absolute value.

Language Objective: I can explain my work using the words distance, absolute value, horizontal distance, and vertical distance.



See **Learn It** for the explanation and worked examples, and the **Vocab** tab for the words. These are your notes to fill in and keep.



Fill in each blank as we go. Use the Word Bank to help you.



WORD BANK — FILL EACH BLANK WITH THE BEST WORD

Distance on the Coordinate Plane

Distance

Absolute value

Horizontal distance

Vertical distance

Coordinate plane

Integer



Tap any word to see what it means and a picture.

Level 3: try the blanks from memory first, then check the bank.

1

— The positive space between points, found from the absolute difference of coordinates on a horizontal or vertical line.

2

How far apart two points are on the coordinate plane is the

3

The distance of a number from zero, used to find distance between points, is its

4

The distance between two points along the x-direction is the

5 The distance between two points along the y-direction is the

6 The grid made by the x-axis and y-axis is the .

7 A whole number or its opposite is an .

Write About the Math

Guided ESOL writing

Use the support level you need. Every level answers the same math question.

1. Understand the Question

EXPLAIN

How does the drone use coordinate distance to calculate the total flight path?

Your job: Answer the question. Use math evidence. Explain how the evidence proves your answer.

Responde la pregunta. Usa evidencia matemática. Explica cómo la evidencia demuestra tu respuesta.

2. Plan Your Math Words

Check at least two words you will use.

- ☐ **Distance on the Coordinate Plane** — The positive space between points, found from the absolute difference of coordinates on a horizontal or vertical line.

Distancia en el plano de coordenadas — El espacio positivo entre puntos, que se halla con la diferencia absoluta de las coordenadas en una línea horizontal o vertical.

- ☐ **Distance** — How far apart two points are. It is never negative.

Distancia — Qué tan separados están dos puntos. Nunca es negativo.

- ☐ **Absolute value** — How far a number is from zero. It is never negative.

Valor absoluto — Qué tan lejos está un número de cero. Nunca es negativo.

- ☐ **Horizontal distance** — How far apart two points are going left or right.

Distancia horizontal — Qué tan separados están dos puntos yendo a la izquierda o a la derecha.

- ☐ **Vertical distance** — How far apart two points are going up or down.

Distancia vertical — Qué tan separados están dos puntos yendo hacia arriba o hacia abajo.

Say it first: Say your idea to a partner or quietly to yourself before you write.

Di tu idea a un compañero o en voz baja antes de escribir.

The horizontal distance from $(-4, 0)$ to $(3, 0)$ is ____ units because ____.

The vertical distance from $(3, 0)$ to $(3, -5)$ is ____ units because ____. **The total flight path is ____ units.**

Di tu respuesta primero: Mi respuesta es ____ porque ____.

3. Build Your Explanation

Model the parts: This model shows the parts of an explanation. It does not answer your writing question.

Claim: I can find the distance by looking at the coordinates that ____.

Evidence: Student finds the distance by subtracting the x-coordinates ($7 - 2 = 5$) since the points share a horizontal line.

Reasoning: This evidence connects the definition of distance on the coordinate plane to the mathematical claim.

Start

Most language support

Complete the frames. Use the word bank.

Completa las oraciones. Usa el banco de palabras.

- The horizontal distance from $(-4, 0)$ to $(3, 0)$ is ____ units because _____. The vertical distance from $(3, 0)$ to $(3, -5)$ is ____ units because _____. The total flight path is ____ units.

- My answer is _____.

Mi respuesta es _____.

Build

Some language support

Write two connected sentences. Use because, so, or but.

Escribe dos oraciones conectadas. Usa porque, entonces o pero.

- My claim is _____.

Mi afirmación es _____.

- I know because _____.

Lo sé porque _____.

- So _____.

Entonces _____.

Explain

Light language support

Write a claim, give math evidence, and explain your reasoning.

Escribe una afirmación, da evidencia matemática y explica tu razonamiento.

- **My claim is ____.**
Mi afirmación es ____.
 - **My math evidence is ____.**
Mi evidencia matemática es ____.
 - **This proves ____ because ____.**
Esto demuestra ____ porque ____.
-
-
-

4. Check Your Explanation

- ☐ I answered the question.
Respondí la pregunta.
- ☐ I used math evidence: numbers, an equation, a model, or a comparison.
Usé evidencia matemática: números, una ecuación, un modelo o una comparación.
- ☐ I explained how the evidence proves my answer.
Explicué cómo la evidencia demuestra mi respuesta.
- ☐ I used at least two math words.
Usé por lo menos dos palabras matemáticas.
- ☐ I reread complete sentences and punctuation.
Volví a leer mis oraciones completas y la puntuación.

